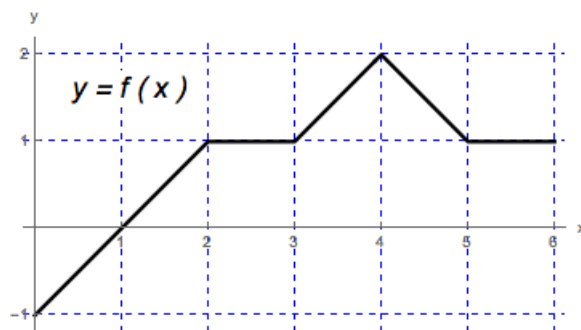


Problem 5

Problem 5

Problem 1 The graph of a function f defined on the interval $[0, 6]$ is given in the figure.



- Compute the net area of the region between the graph of f and the x -axis, on the interval $[0, 6]$.
- Draw a rectangle with base on the x -axis, $0 \leq x \leq 6$, whose area is equal to the net area in part (a).
- In the figure, mark a point c in $(0, 6)$ such that $f(c)$ is the height of the rectangle from part (b).
- Using the figure and parts (a-c), what is the relationship between the rectangle from part (b), the net area from part (a), and the average value of f on $[0, 6]$?